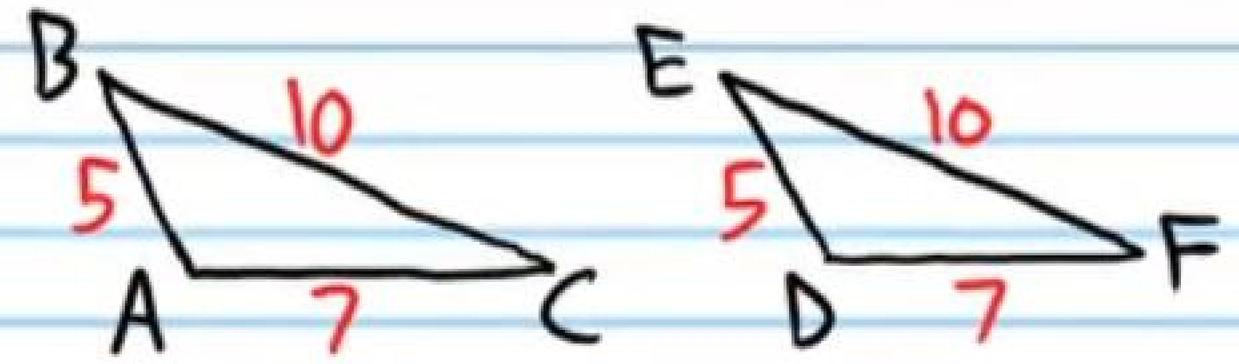
Name
Date
Course

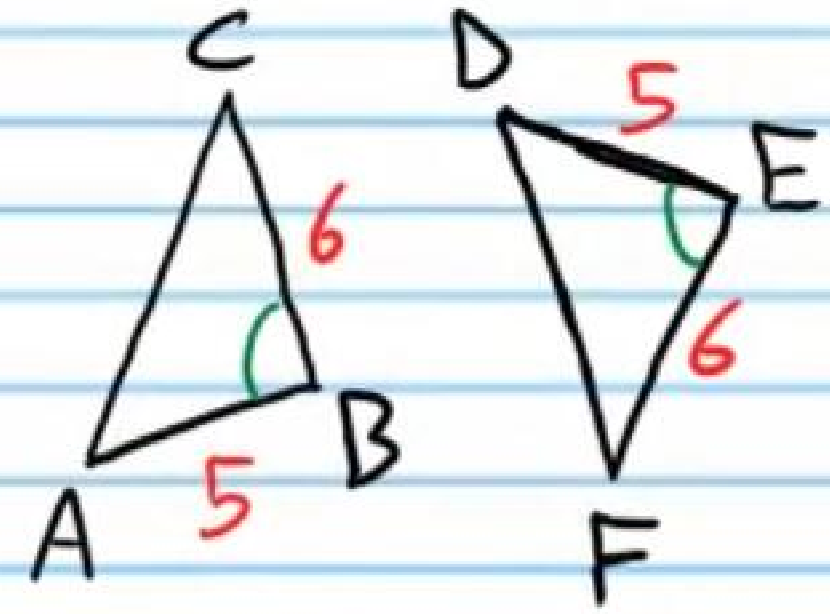
Applications of Triangle Congruence and More Triangle Theorems (Homework).

Write two-column proofs to show that the statements below are true.

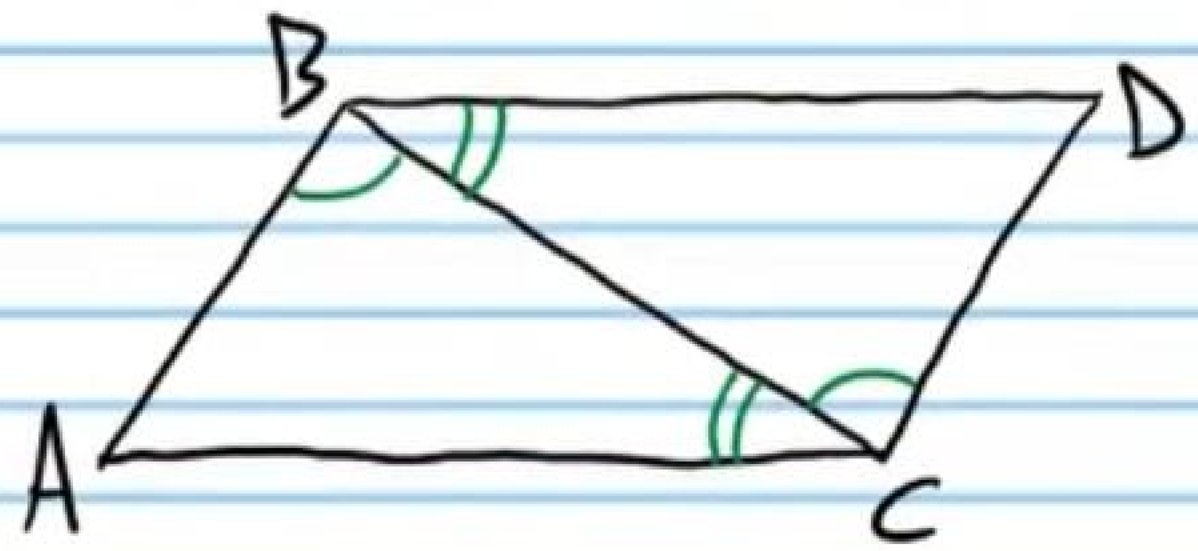
① If $\overline{AB} \cong \overline{DE}$, $\overline{BC} \cong \overline{EF}$, and $\overline{AC} \cong \overline{DF}$ below, then $\angle A \cong \angle D$.



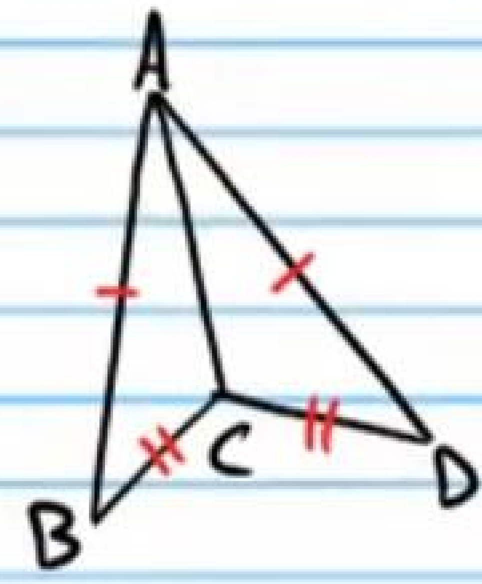
② If $\overline{AB} \cong \overline{DE}$, $\overline{BC} \cong \overline{FE}$, and $\angle B \cong \angle E$, then $\angle C \cong \angle F$.



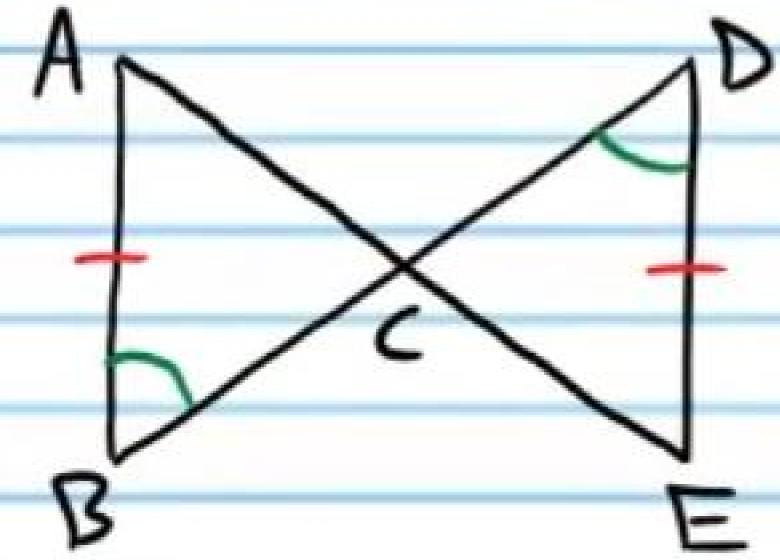
③ If $\angle ABC \cong \angle BCD$ and $\angle ACB \cong \angle CBD$, then $\overline{AB} \cong \overline{DC}$.



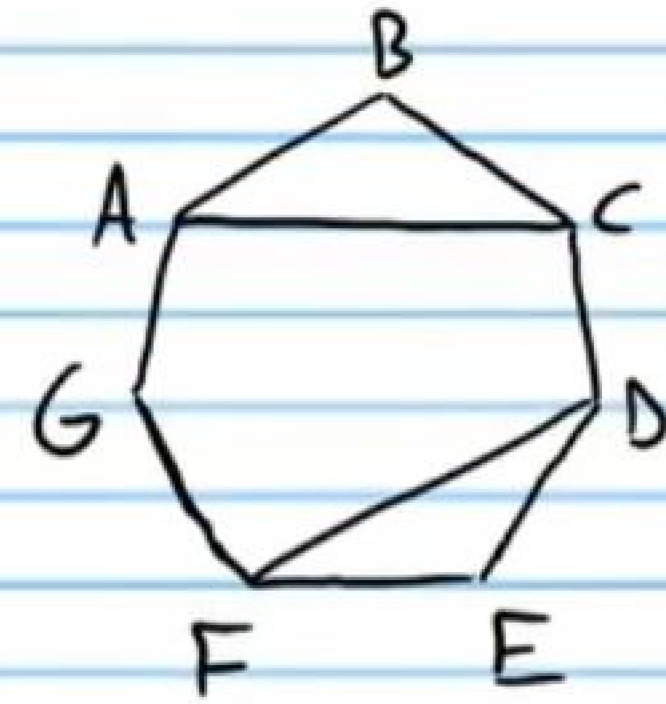
④ If $\overline{AB} \cong \overline{AD}$, and $\overline{BC} \cong \overline{CD}$, then $\overline{AC}$ bisects $\angle BAD$.



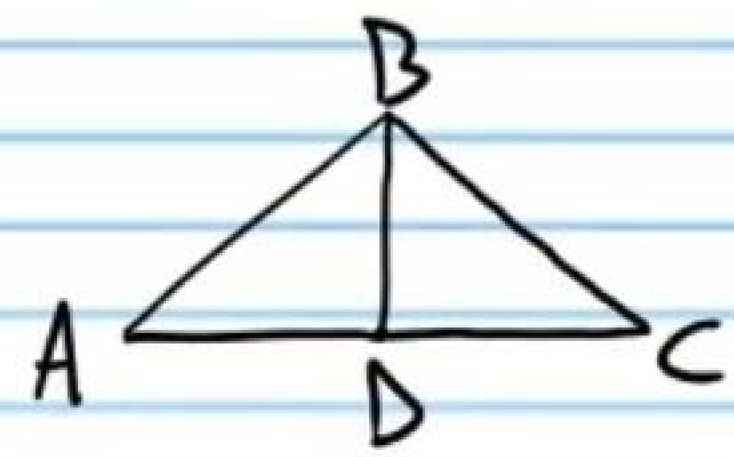
⑤ If $\angle B \cong \angle D$ and $\overline{AB} \cong \overline{DE}$, then C bisects $\overline{AE}$.



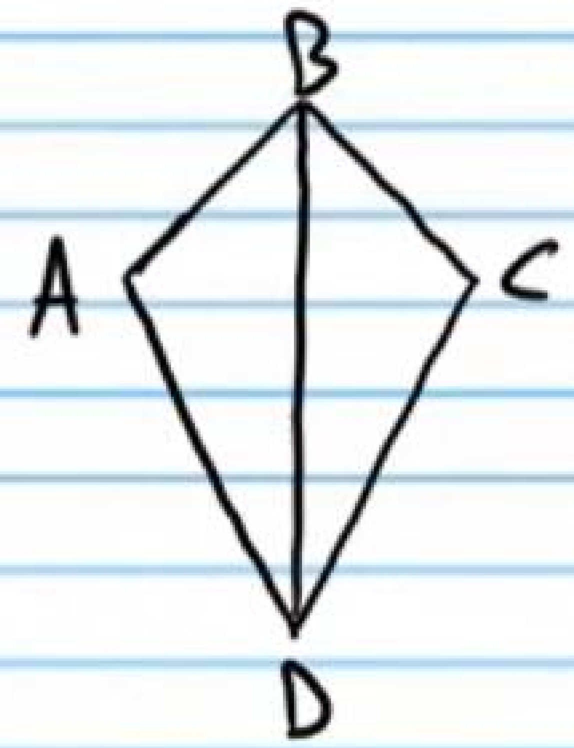
⑥ If ABCDEFG below is a regular heptagon, then $\overline{AC} \cong \overline{FD}$.



⑦ If $\overline{BD} \perp \overline{AC}$, and $\angle A \cong \angle C$, then D is the midpoint of $\overline{AC}$.

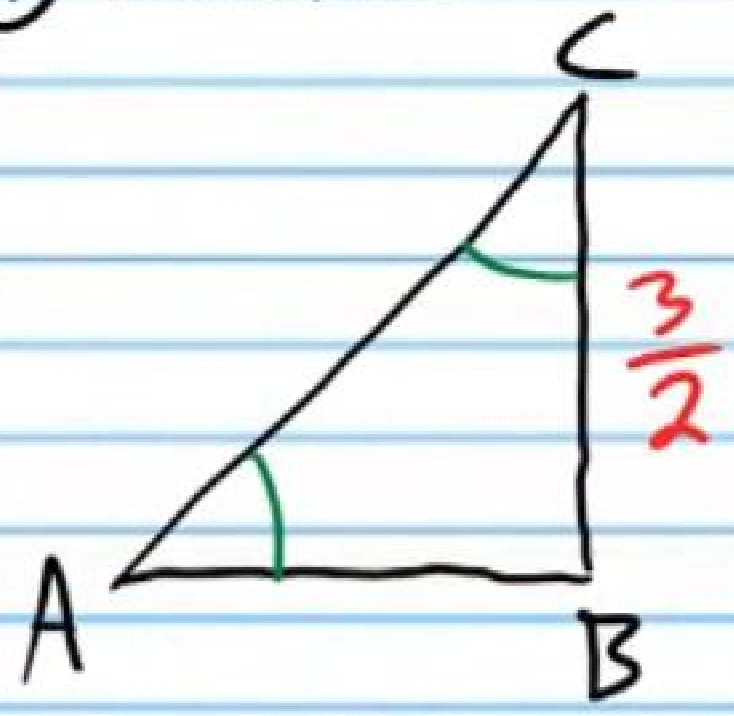


(8) If $\overline{BD}$ bisects $\angle ADC$, and $\angle A \cong \angle C$, then $\overline{AB} \cong \overline{BC}$.



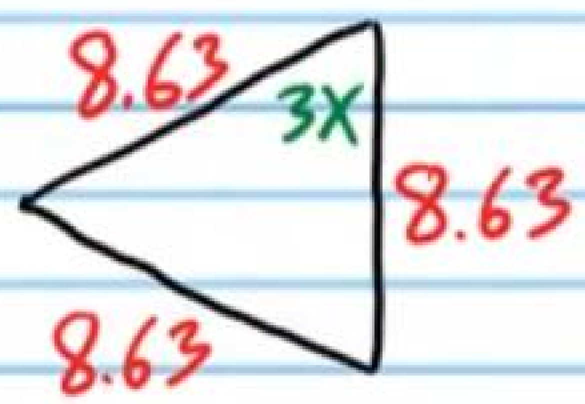
In the following problems, you do not have to write formal proofs.

(9) Find AB



$\boxed{\frac{3}{2}}$

(10) Find x.

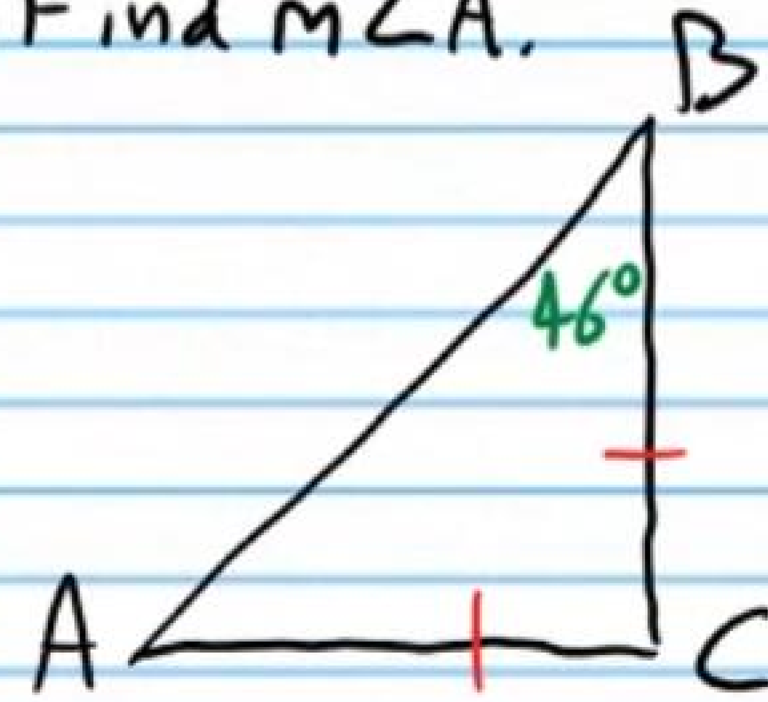


$\boxed{20}$

(11) Find x.

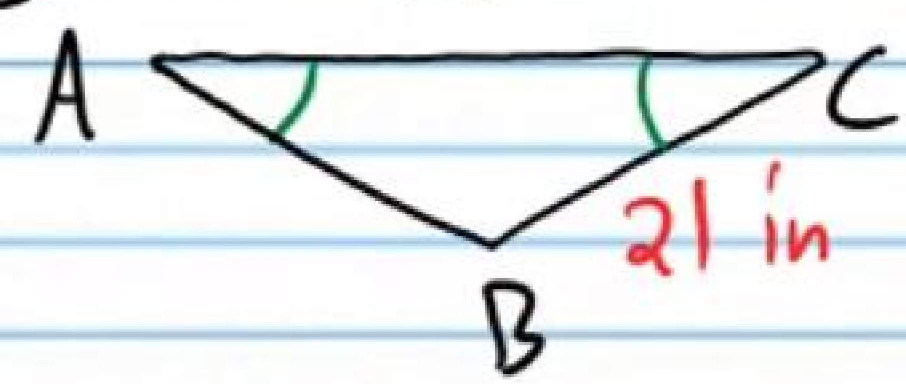
$\boxed{14}$

(12) Find $m\angle A$.



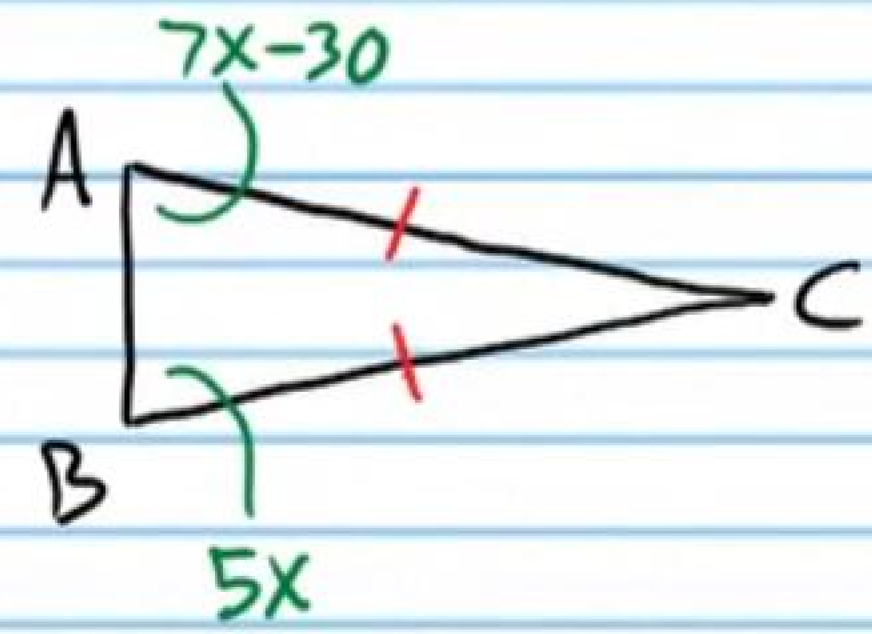
$\boxed{46°}$

(13) Find AB.



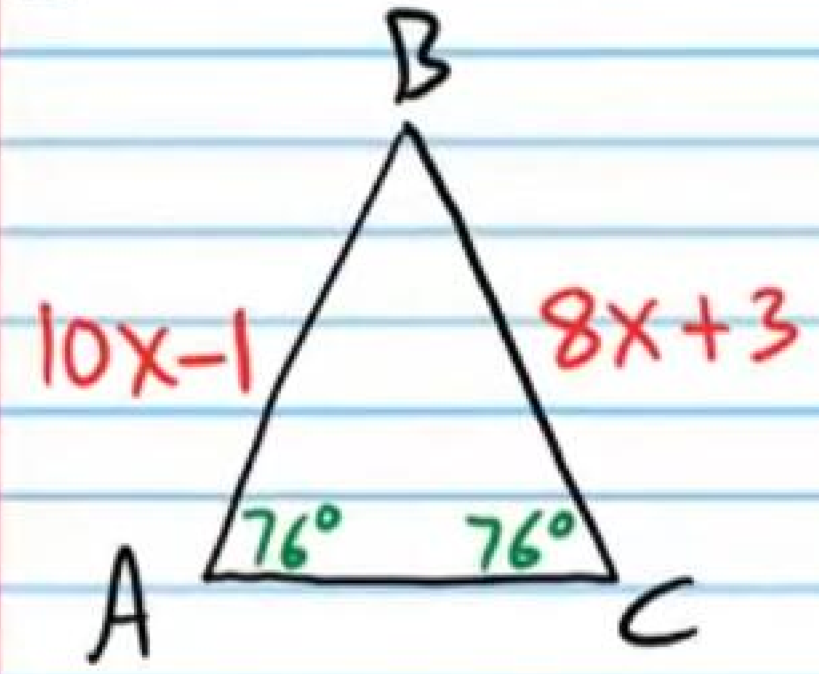
$\boxed{21 \text{ in}}$

(14) Find $m\angle C$.



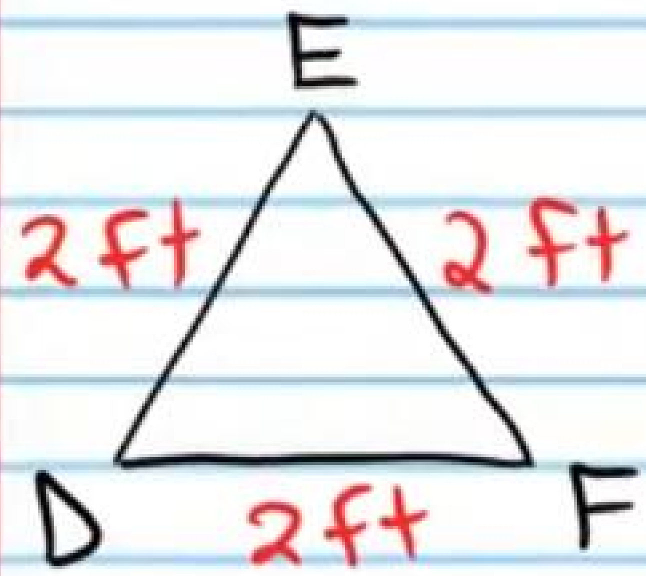
$\boxed{30°}$

(15) Find BC.



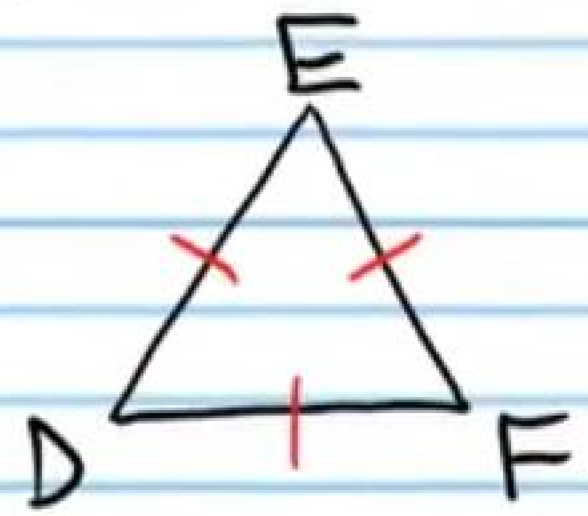
$\boxed{19}$

(16) Find m∠D.



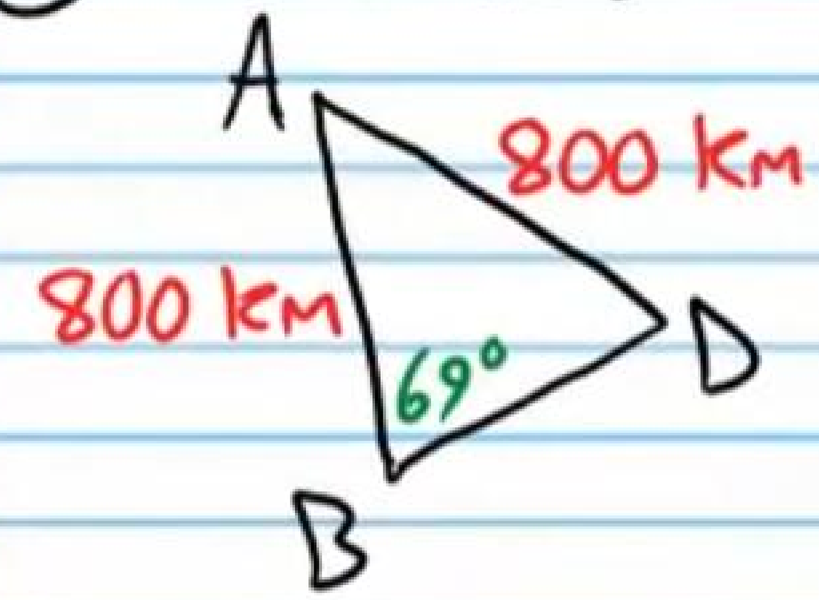
$\boxed{60°}$

(17) Find m∠F.



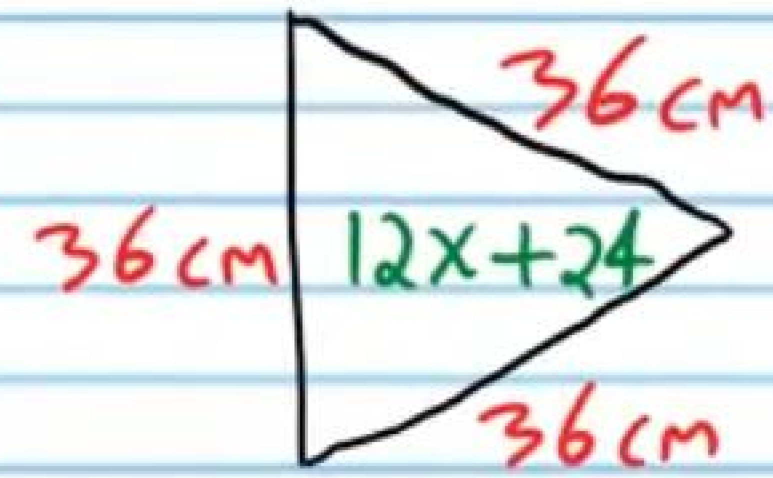
$\boxed{60°}$

(18) Find m∠D.



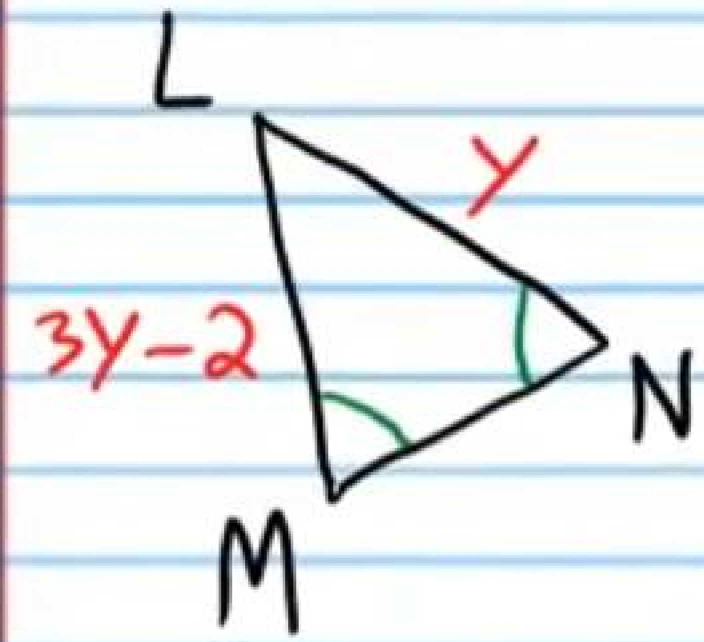
$\boxed{69°}$

(19) Find x.



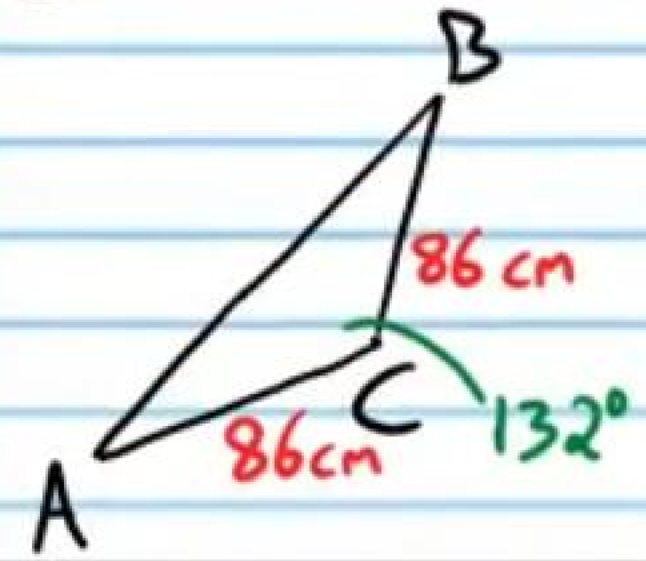
$\boxed{3}$

(20) Find y.



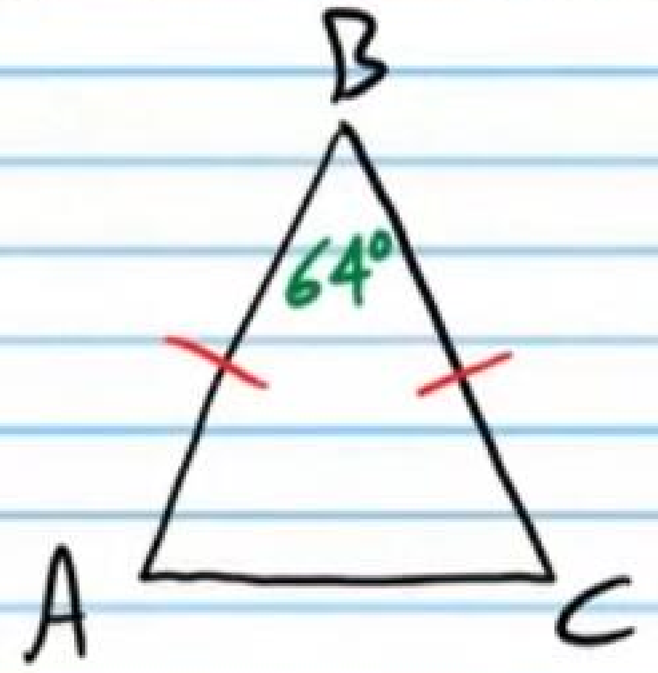
$\boxed{1}$

(21) Find m∠A.



$\boxed{24°}$

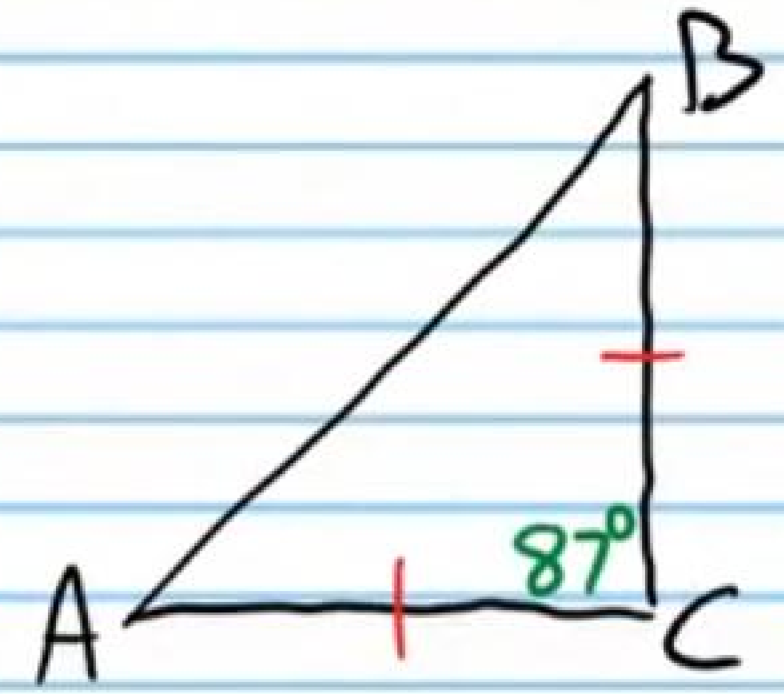
(22) Find $m\angle C$.



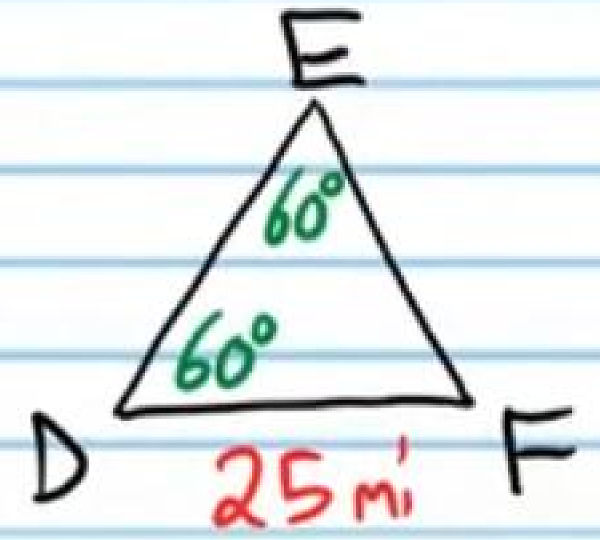
$\boxed{58°}$

Write two-column proofs to show that the following statements are true.

(23) If $m\angle C = 87°$, and $\overline{BC} \cong \overline{AC}$, then $m\angle A = 46.5°$.
Note: You must use the Triangle Sum Theorem.

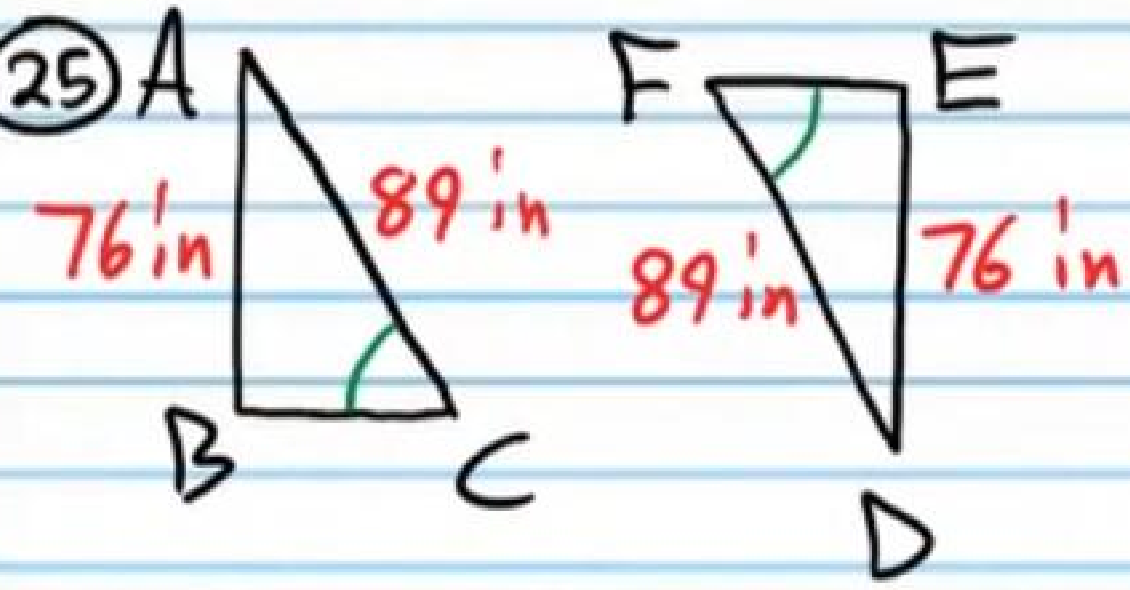


(24) If $m\angle E = 60°$ below, and $m\angle D = 60°$, and $DF = 25$ mi, then $DE = 25$ miles. Note: You must use the Triangle Sum Theorem.



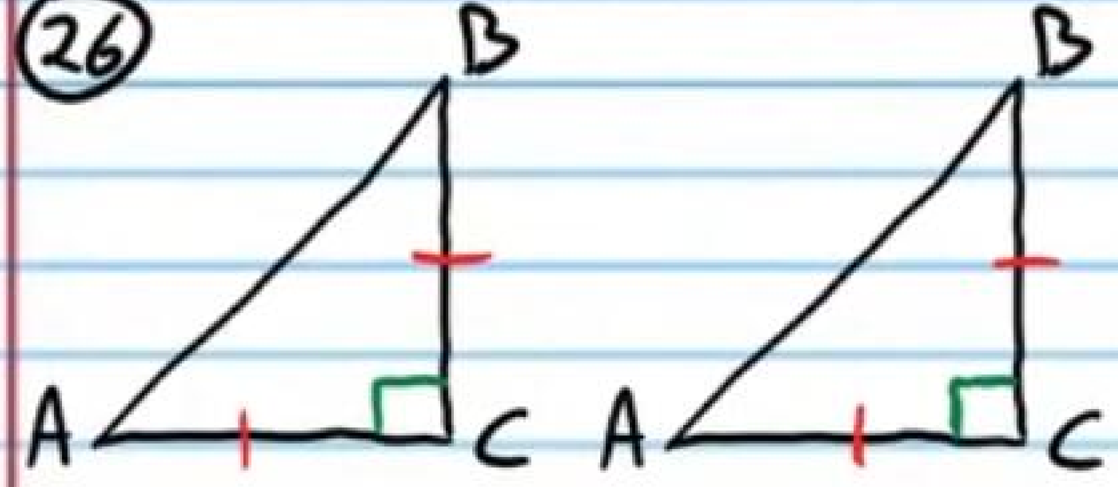
Determine if the triangles below are congruent. If so, state the postulate or theorem used.

(25)



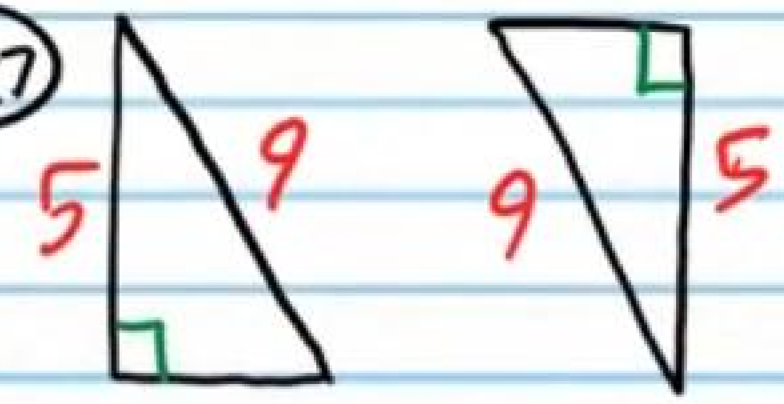
Not enough information

(26)



$\cong$ by SAS.

(27)

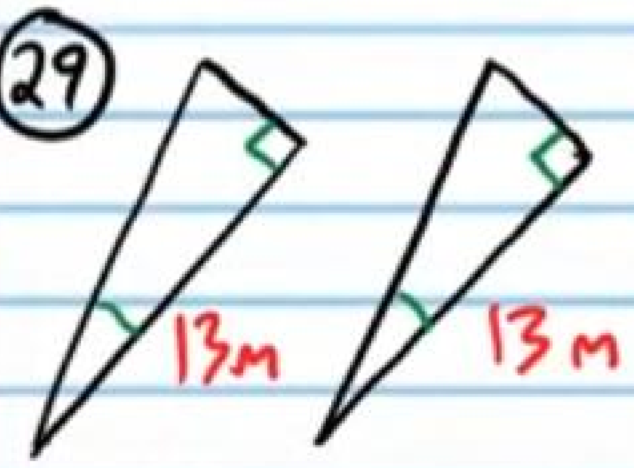


$\cong$ by HL

(28)

$\cong$ by HL

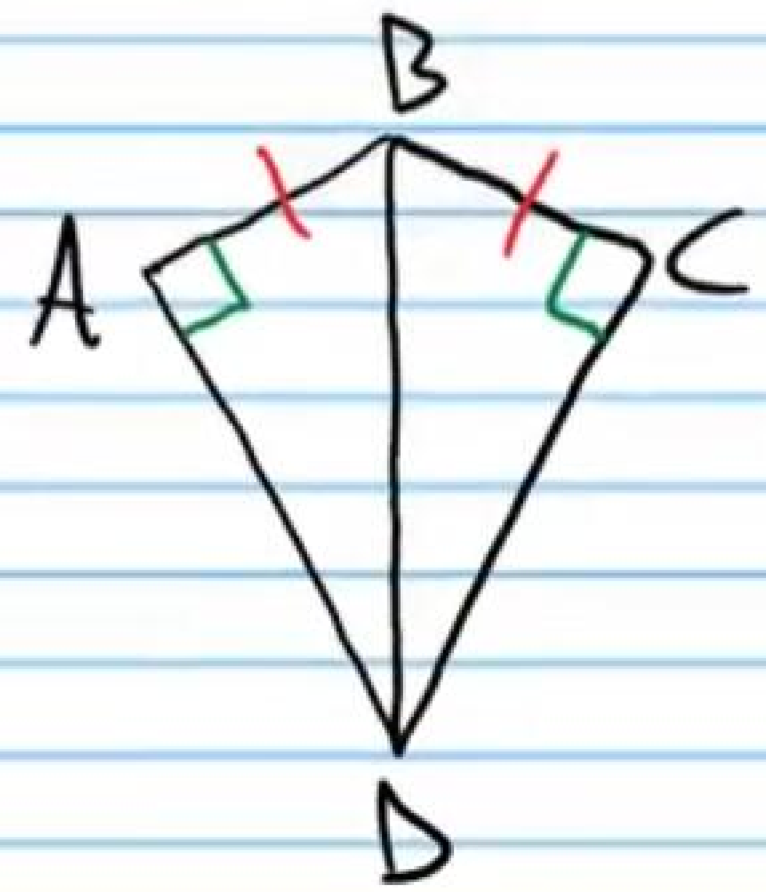
(29)



$\cong$ by ASA

Write a two-column proof to show that the following statement is true.

(30) If $\overline{AB} \cong \overline{BC}$ and $\angle A$ and $\angle C$ are right angles, then $\overline{AD} \cong \overline{CD}$.



Answers for Homework Problems #'s 1–8, 23–24, 30.

①

Statement	Reason
I) $\overline{AB} \cong \overline{DE}$, $\overline{BC} \cong \overline{EF}$, & $\overline{AC} \cong \overline{DF}$	Given.
II) $\triangle ABC \cong \triangle DEF$	SSS
III) $\angle A \cong \angle D$	CPCTC

②

Statement	Reason
I) $\overline{AB} \cong \overline{DE}$, $\overline{BC} \cong \overline{FE}$, and $\angle B \cong \angle E$.	Given.
II) $\triangle ABC \cong \triangle DEF$	SAS
III) $\angle C \cong \angle F$	CPCTC

③

Statement	Reason
I) $\angle ABC \cong \angle BCD$ & $\angle ACB \cong \angle CBD$.	Given.
II) $\overline{BC} \cong \overline{BC}$	Ref. Prop of $\cong$.
III) $\triangle ABC \cong \triangle DCB$	ASA
IV) $\overline{AB} \cong \overline{DC}$	CPCTC

④

Statement	Reason
I) $\overline{AB} \cong \overline{AD}$ & $\overline{BC} \cong \overline{CD}$.	Given.
II) $\overline{AC} \cong \overline{AC}$	Ref. Prop of $\cong$.
III) $\triangle ABC \cong \triangle ADC$	SSS
IV) $\angle BAC \cong \angle DAC$	CPCTC
V) $\overline{AC}$ bisects $\angle BAD$	Def. of $\angle$ bisector.

⑤

Statement	Reason
I) $\angle B \cong \angle D$ & $\overline{AB} \cong \overline{DE}$.	Given.
II) $\angle ACB \cong \angle DCE$	Vert. $\angle$'s Thm.
III) $\triangle ABC \cong \triangle EDC$	AAS
IV) $\overline{AC} \cong \overline{CE}$	CPCTC
V) C bisects $\overline{AE}$.	Def. of Seg bisector.

⑥

Statement	Reason
I) ABCDEFG is a regular heptagon.	Given.
II) $\angle B \cong \angle E$	Def. of regular heptagon.
III) $\overline{AB} \cong \overline{FE}$ & $\overline{BC} \cong \overline{DE}$ or $\overline{AB} \cong \overline{DE}$ & $\overline{BC} \cong \overline{FE}$.	Def. of regular heptagon.
IV) $\Delta ABC \cong \Delta DEF$ or $\Delta ABC \cong \Delta FED$.	SAS
V) $\overline{AC} \cong \overline{FD}$	CPCTC

⑦

Statement	Reason
I) $\overline{BD} \perp \overline{AC}$	Given.
II) $\angle ADB$ & $\angle CDB$ are rt. $\angle$'s.	Def. of $\perp$.
III) $\angle ADB \cong \angle CDB$	Right $\angle \cong$ Thm.
IV) $\angle A \cong \angle C$	Given.
V) $\overline{BD} \cong \overline{BD}$	Ref. Prop of $\cong$.
VI) $\Delta ADB \cong \Delta CDB$	AAS
VII) $\overline{AD} \cong \overline{DC}$	CPCTC
VIII) D is the midpoint of $\overline{AC}$.	Def. of midpoint.

⑧

Statement	Reason
I) $\overline{BD}$ bisects $\angle ADC$.	Given.
II) $\angle ADB \cong \angle CDB$	Def. of $\angle$ bisector.
III) $\angle A \cong \angle C$	Given.
IV) $\overline{BD} \cong \overline{BD}$	Ref. Prop of $\cong$.
V) $\Delta ABD \cong \Delta CBD$	AAS
VI) $\overline{AB} \cong \overline{BC}$	CPCTC

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Statement	Reason
I) $m\angle C = 87°$	Given.
II) $\overline{BC} \cong \overline{AC}$	Given.
III) $m\angle A = m\angle B$	Isosceles Δ Thm.
IV) $m\angle A + m\angle B + m\angle C = 180°$	Δ Sum Thm.
V) $m\angle A + m\angle A + 87 = 180$	Subs. Prop of =.
VI) $2m\angle A + 87 = 180$	Simplification.
VII) $2m\angle A = 93$	Subt. Prop of =.
VIII) $m\angle A = 46.5°$	Div. Prop of =.

24

Statement	Reason
I) $m\angle E = 60°$ & $m\angle D = 60°$	Given.
II) $m\angle D + m\angle E + m\angle F = 180°$	Δ Sum Thm.
III) $60 + 60 + m\angle F = 180°$	Subs. Prop of =.
IV) $120 + m\angle F = 180°$	Simplification.
V) $m\angle F = 60°$	Subt. Prop of =.
VI) ΔDEF is equiangular	Def. of equiangular.
VII) $DF = DE$	Equilateral Δ Conv. Thm.
VIII) $DF = 25$ mi	Given.
IX) $DE = 25$ mi	Trans. Prop of =.

30

Statement	Reason
I) $\overline{AB} \cong \overline{BC}$	Given
II) $\angle A$ & $\angle C$ are right $\angle$'s.	Given.
III) $\angle A \cong \angle C$	Rt $\angle \cong$ Thm.
IV) $\overline{BD} \cong \overline{BD}$	Ref. Prop of $\cong$.
V) $\Delta ABD \cong \Delta CBD$	HL
VI) $\overline{AD} \cong \overline{CD}$	CPCTC